

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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*Dedicated to my family who endured missed birthdays,
missed festivals and countless days alone so that I may
make history*

0.1 Datasets, Simulation Samples and Theoretical Predictions

0.1.1 Datasets

0.2 Datasets

0.2.1 Data

Triggers and Datasets

This analysis uses a data sample recorded by the CMS experiment during 2016, 2017 and 2018, corresponding to 137.1 fb^{-1} of data.

The datasets used are listed and documented in ???. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. To avoid duplicate events from different primary datasets, events are taken:

- from EGamma if they pass the diEle or triEle or singleElectron triggers,
- from DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- from MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- from SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for 2018 collision data is listed in ???.

Trigger Efficiency

The efficiency in data of the combination of triggers used in the analysis with respect to the offline reconstruction and selection is measured by considering 4ℓ events triggered by single lepton triggers. One of the four reconstructed leptons (the “tag”) is geometrically matched to a trigger object passing the final filter of one of the single muon or single electron triggers. The other three leptons are used as “probes”. In each 4ℓ event there are up to 4 possible tag-probe combinations, and all possible combinations are counted in the denominator of the efficiency. For each of the three probe leptons all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the analysis. If any of the analysis triggers can be formed using the probe leptons, the set of probes is also counted in the numerator of the efficiency.

This method does not have a perfect closure in MC events due to the fact that the presence of a fourth lepton increases the trigger efficiency, and this effect is

not accounted for. Also, in the $2e2\mu$ final state, the three probe leptons cannot be combined to form all possible triggers which can collect events with two electrons and two muons (e.g. if the tag lepton is an electron, the three remaining leptons cannot pass a double electron trigger). Therefore the method is also exercised on MC and the difference between data and MC is used to determine the reliability of the simulation. The efficiency plotted as a function of the minimum p_T of the three probe leptons in data and MC using this method can be seen in Fig. ?? that shows the trigger efficiency in 2018 data for different final states. The MC efficiency describes well the data within the statistical uncertainties.

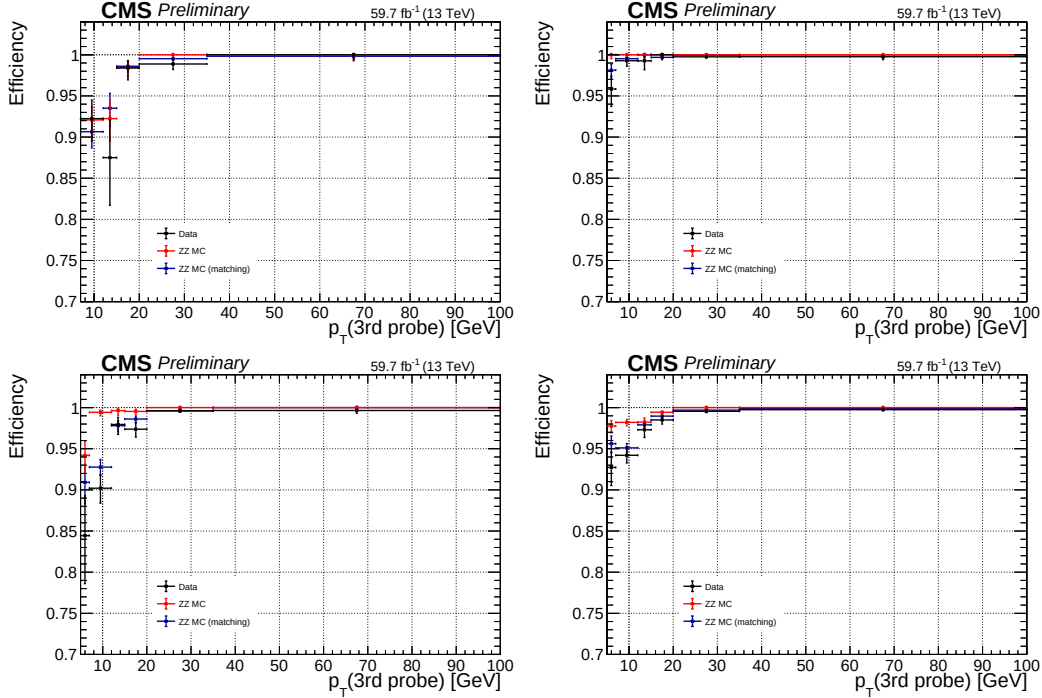


FIGURE 1: Trigger efficiency measured in 2018 data using 4ℓ events collected by single lepton triggers for the $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

0.2.2 Simulation Samples and Theoretical Predictions

The SM Higgs boson production by gluon fusion ($gg \rightarrow H$) process descriptions (Figure) are acquired using the HRES [deFlorian:2012mx, Grazzini:2013mca], POWHEG MINLO-HJ [Hamilton:2012np] and POWHEG V2 [powhcg, powhegvv] generators. The $gg \rightarrow H$ process description at next-to-next-to-leading order (NNLO) accuracy in perturbative QCD and next-to-next-to-leading logarithmic (NNLL) accuracy in resummation of soft-gluon effects at small transverse momenta is obtained by partonic level MC generator HRES. The inclusive jet associated activity predictions of fiducial cross section is attained by HRES because the resummation is

inclusive over the QCD radiation recoiling against the Higgs boson. HRES predictions are obtained using the central values for the factorization and normalization scales to be $m_H = 125$ GeV. The partonic level matrix-element generator, the POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with an connection to parton-shower programs. The Higgs boson production via $gg \rightarrow H$ with zero associated jets can be described at NLO accuracy using the POWHEG. For this analysis, the p_T spectrum of the Higgs boson obtained by POWHEG has been reweighted to match closely with HRES generator in the full phase space. This factor shrinks additional jets emission in large p_T region, and raises the contribution in the region where p_T approaches to zero by the Sudakov form factor. The $gg \rightarrow H$ production in association with up to one additional jets is improved to next-to-leading logarithmic (NLL) accuracy by extending POWHEG generator to the POWHEG MINLO-HJ generator based on the MiNLO prescription [Hamilton:2012np]. The POWHEG MINLO-HJ generator provides NLO accuracy $gg \rightarrow H$ production in association with zero and one additional jets and LO accuracy for two jets. Finite masses of top and bottom quarks has been considered for $gg \rightarrow H$ production for all generators. The SM Higgs boson production by

Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [Alioli:2008gx, Nason:2004rx, Frixione:2007vw] generator for the five main production modes: gluon fusion ($gg \rightarrow H$) including quark mass effects [Bagnaschi:2011tu], vector boson fusion (VBF) [Nason:2009ai], and associated production (WH, H and H [Hartanto:2015uka]). In the case of WH and H the MINLO HVJ extension of POWHEG is used [Luisoni:2013kna]. The description of the decay of the Higgs boson to four leptons is obtained using the JHUGEN generator [Gao:2010qx]. In the case of WH, H and H, the Higgs boson is allowed to decay to $H \rightarrow \gamma\gamma$ such that 4-lepton events where two leptons originate from the decay of associated, W bosons or top quarks are also taken into account in the simulation. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the NNPDF 3.1 NLO parton distribution functions (PDFs) [Ball:2014uwa]. The list of signal samples and their cross sections are shown in ???. A subset of these samples are used to asses the model dependence of the measurement results are indicated in the Table 2.

The MC samples generated by POWHEG and PYTHIA8.209 are used to extract fiducial phase space acceptance and reconstruction efficiencies for individual models separately as described in Section 0.1.1. These samples, along with the samples generated by using the HRES and the POWHEG MINLO-HJ with alternate representation of $gg \rightarrow H$ process, are utilized for the comparison of the measurement results with theoretical predictions based on SM calculation in Section ??.

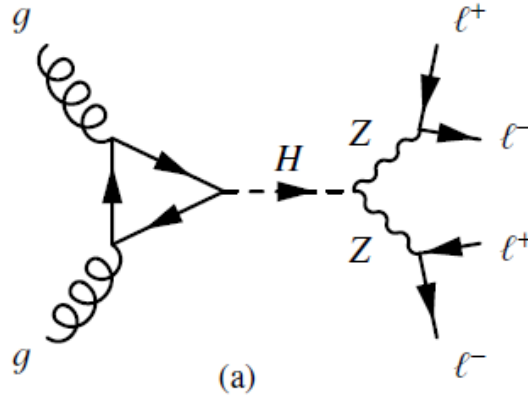


FIGURE 2: $gg \rightarrow H$ process fynman diagram.

All SM production modes have been produced individually to estimate the dependence of the measurements on underlying assumptions of Higgs boson production mechanism. We have also simulated a wide range of samples that provide descriptions for exotic Higgs-like resonances productions and their decay to the four-lepton final state to estimate the dependence of the measurements on the Higgs boson properties. These include spin-zero, spin-one and spin-two resonances with anomalous interactions with a pair of neutral gauge bosons (ZZ , $Z\gamma^*$, $\gamma^*\gamma^*$) described by higher-order operators, as discussed in details in Ref. [Khachatryan:2014kca]. The details of the method used to asses the model dependence will be described in Section ?? . All these MC samples are produced using the POWHEG generator for the representation of NLO QCD effects in the production mechanisms, and JHUGEN [Gao:2010qx, Bolognesi:2012mm, Anderson:2013afp] to represent the decay of these exotic resonances to four-leptons containing all spin correlations.

Background Samples

For the estimations of the backgrounds using MC samples, the background process $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ was generated at NLO accuracy containing s-, t- and u-channel diagrams (Figure ??) using POWHEG V2 [Nason:2013ydw] and PYTHIA8, with the same settings as for the Higgs signal. As this simulation covers a large range of ZZ invariant masses, dynamical QCD factorization and renormalization scales have been chosen, equal to m_{ZZ} .

The $gg \rightarrow ZZ$ process (Figure ??) is simulated at leading order (LO) with MCFM [MCFM, Campbell:2013una]. In order to match the $gg \rightarrow H \rightarrow ZZ$ transverse momentum spectra predicted by POWHEG at NLO, the showering for MCFM samples is performed with different PYTHIA8 settings, allowing only emissions up to the parton-level scale (“wimpy” shower).

The initial- and final-state radiation have been included for all events generators, which can guide to the existence of extra hard photons in the event. The

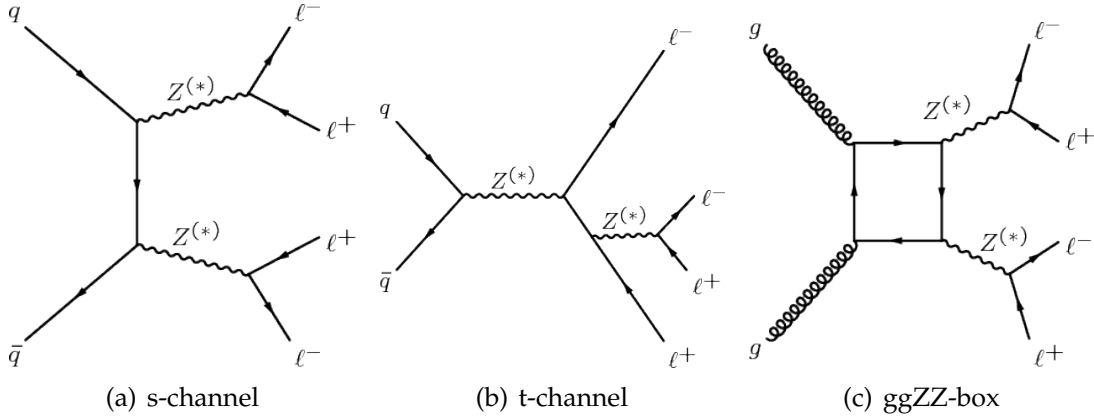


FIGURE 3: $q\bar{q}toZZ \rightarrow 4\ell$ process s- and t- channel fynman diagrams (a) and (b). $gg \rightarrow ZZ \rightarrow 4\ell$ process fynman diagram (c).

POWHEG and JHUGEN event generators also properly take into account interference between all contributing amplitudes of the $H \rightarrow 4\ell$ process, containing those affiliated to the reordering of identical leptons in the 4μ and $4e$ final states. The set of parton distribution functions (PDF) CTEQ6L [cteq66], CT10 [CT10] and MSTW2008 [MSTW08] are used for LO, NLO, and NNLO generators.

To include the fact of parton shower, underlying events and hadronization, all generated events were interfaced to PYTHIA 6.4 Tune $Z2^*$ [Chatrchyan:2011id]. The PYTHIA 6.4 $Z2^*$ tune is obtained from the Z1 tune [Field:2010bc], that takes the CTEQ5L parton distribution set, whereas $Z2^*$ follows CTEQ6L [Pumplin:2002vw]. HRES does not supply an interface with the programs that can include the effects of hadronization and multi-parton interaction. To correct HRES predictions for these effects, the POWHEG+JHUGEN events without hadronization and multi-parton interactions were reweighted to match closely the HRES events in a phase space minimally greater than fiducial phase space. After that, the hadronization and multi-parton interaction effects are included in reweighted POWHEG+JHUGEN events by an interface with PYTHIA. The procedure of reweighting events to match HRES to add non-perturbative effects to HRES partonic level predictions was repeated for all studied observables for integrated and differential cross section measurements.

All the generated events were processed via a full simulation of the CMS detector based on GEANT4 [GEANT] and were reconstructed with the same algorithms that are used for data analysis. Multiple interactions (pileup) are incorporated in simulations by reweighting the number of primary vertex distribution to match the distribution of the observed data. Energy deposits from pileup interactions and underlying events are subtracted when computing the energy of jets and isolation of reconstructed objects using the FASTJET technique [Cacciari:2007fd, Cacciari:2008gn, Cacciari:2011ma]. This technique is based on the calculation of the η -dependent average energy density evaluated individually for each event.

All the data-to-MC scale factors are determined using a “tag and probe” method [CMS:2011lsa], discussed in Sections ?? and ??, and are used to correct simulated

samples to compensate the differences of the leptons selection efficiencies in data and simulation. The dependency of the selection efficiency on the number of reconstructed primary vertices in the event is found to be negligible for data samples. More details are presented and discussed in Ref. [Chatrchyan:2013mx].

Pileup Reweighting

For each year, corresponding simulation samples are reweighted to match the pileup distribution in data. An example of reweighting procedure for 2018 data is shown in Fig. ??.

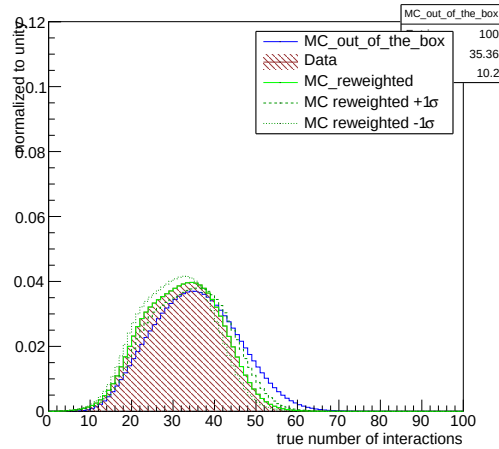


FIGURE 4: Distribution of pileup in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weighs are also shown.

0.3 Analysis Strategy

The integrated and differential cross sections are measured in a fiducial region in order to reduce the effects of model dependent acceptances. The fiducial selection mimics the reconstruction level selection, which is optimized to detect a low mass ($m \sim 125$ GeV) Higgs boson decaying to 4 leptons through two Z bosons. Since in the SM there are multiple production modes, the fiducial selection and measurement strategy are designed to be independent of how the Higgs boson is produced.

For this reason, the inclusion of isolation in the fiducial phase space definition is necessary in order to make the reconstruction efficiency independent with respect to the fiducial phase space independent of the number of jets.

Furthermore, in the sub-leading SM production modes there can be associated vector bosons which can decay to leptons, and therefore the reconstructed Higgs

boson candidate may not be composed of the correct combination of the leptons. In order to retain the full information from an unbinned simultaneous fit of the signal and background pdfs to the four-lepton mass distribution, it is necessary to treat such “wrong combination” events as a background.

Further signal-induced background can be formed due to imperfect correspondence between quantities at the particle level and the corresponding quantities at the reconstruction level. Signal events which are from the fiducial phase space but fail the reconstruction are accounted in the fiducial efficiency (ϵ) and the expected contribution from events which pass the reconstruction selection but are not from within the fiducial phase space (f_{nonfid}) are referred to as “nonfiducial signal” events and are treated as a background, illustrated in Figure ??.

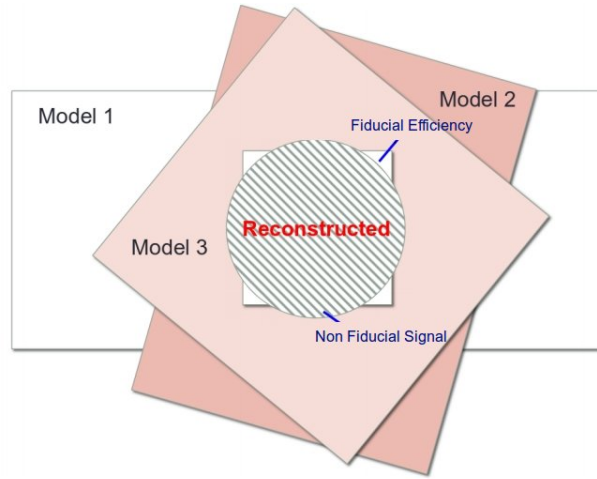


FIGURE 5: Fiducial efficiency(ϵ) and nonfiducial(f_{nonfid}) signal definition.

The effects of imperfect detector resolution can in general have a non-negligible impact on the shape of the distribution of the measured observables. For this reason, in case of the differential cross section measurements, a procedure to correct for the detection efficiencies and resolution effects is applied. Throughout this document, this procedure will be referred to as the unfolding procedure, and the unfolded differential distributions will be referred to as distributions at the fiducial level. The reconstruction level selection, fiducial phase space definition, statistical procedure, and their model dependence will be described in the subsequent Sections.

0.3.1 Event Selection and Background Modeling

Event Selection

The event selection for these measurements are similar to the event selection used in previous Higgs boson properties measurements in the same final state [HIG-18-001,

HIG-16-041, Chatrchyan:2013mx, Khachatryan:2014kca]. The online event selection is based on having fired High Level Trigger paths as described in [sec:trigpaths]. Unlike in the Run I analysis, the trigger requirement does not depend on the selected final state: it is always the OR of all HLT paths. The reason is in Run II we will be targeting associated production modes that can come with additional leptons, thus improving trigger efficiency further. The events are required to have at least one good primary vertex (PV) fullfilling the following criteria: high number of degree of freedom ($N_{PV} > 4$), collisions restricted along the z -axis ($z_{PV} < 24$ cm) and small radius of the PV ($r_{PV} < 2$ cm).

The minimum p_T of 17 and 8 GeV is required for leading and subleading leptons for double leptons triggers, while 15, 8 and 5 GeV for triple electron trigger.

0.3.2 ZZ Candidate Selection

The four-lepton candidates are built from what we call **selected leptons**, which are the tight leptons (defined in sections ?? and ??) that pass the $SIP_{3D} < 4$ vertex constraint and the isolation cuts (defined in sections ?? and ??), where FSR photons are subtracted as described in Section ?. A lepton cross cleaning is applied by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are defined as pairs of selected leptons of opposite charge and matching flavour (e^+e^- , $\mu^+\mu^-$) that satisfy $12 < m_{\ell\ell(\gamma)} < 120$, where the Z candidate mass includes the selected FSR photons if any.
2. **ZZ candidates** are defined as pairs of non-overlapping Z candidates. The Z candidate with reconstructed mass $m_{\ell\ell}$ closest to the nominal Z boson mass is denoted as Z_1 , and the second one is denoted as Z_2 . ZZ candidates are required to satisfy the following list of requirements:
 - **Ghost removal** : $\Delta R(\eta, \phi) > 0.02$ between each of the four leptons.
 - **lepton p_T** : Two of the four selected leptons should pass $p_{T,i} > 20$ and $p_{T,j} > 10$.
 - **QCD suppression**: all four opposite-sign pairs that can be built with the four leptons (regardless of lepton flavor) must satisfy $m_{\ell\ell} > 4$. Here, selected FSR photons are not used in computing $m_{\ell\ell}$, since a QCD-induced low mass dilepton (eg. J/Ψ) may have photons nearby (e.g. from π_0).
 - **Z_1 mass**: $m_{Z_1} > 40$
 - **'smart cut'**: defining Z_a and Z_b as the mass-sorted alternative pairing Z candidates (Z_a being the one closest to the nominal Z boson mass), require NOT($|m_{Z_a} - m_Z| < |m_{Z_1} - m_Z|$ AND $m_{Z_b} < 12$). Selected FSR photons are included in m_Z 's computations. This cut discards 4μ and $4e$ candidates where the alternative pairing looks like an on-shell Z + low-mass $\ell^+\ell^-$. (NB. In Run I, such a situation was avoided by choosing the

best ZZ candidate before applying kinematic cuts to it, most precisely before the $m_{Z2} > 12$ cut. The present smart cut allows to choose the best ZZ candidate after all kinematic cuts.)

- **four-lepton invariant mass:** $m_{4\ell} > 70$

3. Events containing at least one selected ZZ candidate form the **signal region**.

0.3.3 Choice of the best ZZ Candidate

Unlike in the Run I analysis, the best ZZ candidate is now chosen after all kinematic cuts, a change that allows to test other selection strategies for this candidate choice. This is especially relevant for events with more than four selected leptons, having in mind the search for the Higgs boson production modes where associated particles can decay to leptons, such as VH and ttH.

For fiducial cross section measurement, Z_1 is chosen with mass closest to nominal Z boson mass and Z_2 is chosen from the Z candidates whose lepton pair gives higher p_T sum. For STXS measurement, we adopt a different approach: if more than one ZZ candidate survives the selection, we choose the one with the highest value of $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ (defined in Section ??, except when two candidates consist of the same four leptons in which case the candidate with Z_1 closest in mass to nominal Z boson mass is retained).

Reconstruction of the jets is incumbent to explore the differential cross section measurements with respect to jets related observables. Jets are reconstructed with $p_T \geq 30$ GeV and $|\eta| \leq 4.7$, discussed in Section ??. To remove jets overlap with leptons or selected photons, it is required to have at least $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} > 0.5$ [Chatrchyan:2013mx] from any of the selected leptons and FSR photons candidates.

Background Modeling

The dominant and sub-dominant irreducible backgrounds for $H \rightarrow 4\ell$ process are ZZ production via $q\bar{q}$ annihilation and gg fusion respectively. The reducible background is of the instrumental nature and arises from the process where partial jets are misidentified as leptons. The main processes which contributes in this background are: jets production in association with Z boson, jets production in association with WZ boson pair and the $t\bar{t}$ pair production. From now on, this background will be denoted as Z+X.

The irreducible backgrounds $q\bar{q} \rightarrow ZZ$ and $gg \rightarrow ZZ$ for $H \rightarrow 4\ell$ integrated and differential measurements are estimated from Monte Carlo (MC) simulation as done earlier in Ref. [Chatrchyan:2013mxa]. In addition, following the study in Ref. [Bonvini:2013], in these measurements we have increased the expected yield for the $gg \rightarrow ZZ$ background by assigning to the leading order (LO) background cross section a $m_{4\ell}$ dependent k-factor equal to the one used for the signal $H \rightarrow 4\ell$ process. In the region ($105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of the expected $gg \rightarrow ZZ$ background by a factor of ~ 2.66 . After the inclusion of this

k-factor, the $gg \rightarrow ZZ$ process is still a sub-dominant background in all $H \rightarrow 4\ell$ measurements.

The reducible background (Z+X) is evaluated using a tight-to-loose “fake rate” method and the control regions in the data, following Ref. [Chatrchyan:2013mxa]. The yields of the Z+X background are determined by two methods (opposite sign (OS) or same sign (SS)) are found to be consistent with each within statistical uncertainties. The first control region is obtained by requiring that two leptons passing loose selections which does fulfill the criteria of Z_1 candidate (failing ID and isolation requirements), while the other two leptons in the event passes the criteria of Z_1 candidate. The sample is represented as “2 Prompt + 2 Fail” (2P2F) sample. This sample is populated with the events contains only two prompt leptons (mostly

TABLE 1: The number of observed candidates, expected background and signal events passing all the selections in the mass range of $m_{4\ell} > 70$ GeV corresponding to three different years. ZZ and signal events are estimated using MC simulations while Z+X is estimated by control regions in data.

Channel	4e	4 μ	2e2 μ	4 ℓ
2016 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	$192.7^{+18.6}_{-20.1}$	$360.2^{+24.9}_{-27.3}$	$471.0^{+32.6}_{-35.7}$	$1023.9^{+68.9}_{-76.0}$
$gg \rightarrow ZZ$	$41.2^{+6.3}_{-6.1}$	$69.0^{+9.5}_{-9.0}$	$101.7^{+14.0}_{-13.3}$	$211.8^{+28.9}_{-27.5}$
Z + X	$21.1^{+8.5}_{-10.4}$	$34.4^{+14.5}_{-13.2}$	$59.9^{+27.1}_{-25.0}$	$115.4^{+31.9}_{-30.1}$
Sum of backgrounds	$255.0^{+23.9}_{-25.1}$	$463.5^{+31.9}_{-33.7}$	$632.6^{+44.2}_{-46.1}$	$1351.1^{+85.8}_{-91.2}$
Signal ($m_H = 125$ GeV)	$12.1^{+1.3}_{-1.4}$	23.8 ± 2.1	30.3 ± 2.6	66.3 ± 5.6
Total expected	$267.1^{+24.9}_{-26.1}$	$487.4^{+33.1}_{-34.9}$	$662.9^{+45.7}_{-47.5}$	$1417.4^{+89.1}_{-94.3}$
Observed	293	505	681	1479
2017 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	$233.86^{+y.y}_{-z.z}$	$441.08^{+y.y}_{-z.z}$	$569.51^{+y.y}_{-z.z}$	$1244.45^{+y.y}_{-z.z}$
$gg \rightarrow ZZ$	$49.11^{+y.y}_{-z.z}$	$81.50^{+y.y}_{-z.z}$	$121.05^{+y.y}_{-z.z}$	$251.66^{+y.y}_{-z.z}$
Z + X	$12.39^{+y.y}_{-z.z}$	$36.39^{+y.y}_{-z.z}$	$44.45^{+y.y}_{-z.z}$	$93.24^{+y.y}_{-z.z}$
Sum of backgrounds	$295.37^{+y.y}_{-z.z}$	$558.97^{+y.y}_{-z.z}$	$735.02^{+y.y}_{-z.z}$	$1589.35^{+y.y}_{-z.z}$
Signal ($m_H = 125$ GeV)	$13.85^{+y.y}_{-z.z}$	$28.64^{+y.y}_{-z.z}$	$35.55^{+y.y}_{-z.z}$	$78.04^{+y.y}_{-z.z}$
Total expected	$309.22^{+y.y}_{-z.z}$	$587.60^{+y.y}_{-z.z}$	$770.57^{+y.y}_{-z.z}$	$1667.39^{+y.y}_{-z.z}$
Observed	307	602	797	1706
2018 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	$333.01^{+y.y}_{-z.z}$	$622.20^{+y.y}_{-z.z}$	$815.20^{+y.y}_{-z.z}$	$1770.41^{+y.y}_{-z.z}$
$gg \rightarrow ZZ$	$75.11^{+y.y}_{-z.z}$	$116.55^{+y.y}_{-z.z}$	$176.86^{+y.y}_{-z.z}$	$368.52^{+y.y}_{-z.z}$
Z + X	$19.37^{+y.y}_{-z.z}$	$50.75^{+y.y}_{-z.z}$	$64.84^{+y.y}_{-z.z}$	$134.96^{+y.y}_{-z.z}$
Sum of backgrounds	$427.50^{+y.y}_{-z.z}$	$789.50^{+y.y}_{-z.z}$	$1056.89^{+y.y}_{-z.z}$	$2273.89^{+y.y}_{-z.z}$
Signal ($m_H = 125$ GeV)	$19.58^{+y.y}_{-z.z}$	$40.83^{+y.y}_{-z.z}$	$50.68^{+y.y}_{-z.z}$	$111.09^{+y.y}_{-z.z}$
Total expected	$447.08^{+y.y}_{-z.z}$	$830.33^{+y.y}_{-z.z}$	$1107.57^{+y.y}_{-z.z}$	$2384.98^{+y.y}_{-z.z}$
Observed	462	850	1130	2442

Drell-Yan with small fraction of Z/γ^* and $t\bar{t}$ events). The second control region is obtained by requiring that three leptons pass the final requirements. The sample is represented as "3 Prompt + 1 Fail" (3P1F) sample.

The $m_{4\ell}$ shape of the reducible background is attained from OS method by performing a fit of $m_{4\ell}$ distributions of 3P1F and 2P2F events independently by empirical analytical functional constructed by Landau [landau:1944] in Ref. [Chatrchyan:2013mx]. Prediction for the reducible background in all three channels together fitted using this empirical shape with indicated total uncertainty is shown in Figure ?? . In case of the differential $H \rightarrow 4\ell$ measurements, the shape of the reducible background is obtained from the templates built from the control regions in data for each bin of the observable considered for the measurement, following a procedure similar to the one used in the spin-parity studies described in the Ref. [Chatrchyan:2013mx]. More details on this procedure and template shapes for each of the differential $H \rightarrow 4\ell$ measurements will be given in Section ?? .

The number of estimated signal and background events, along with observed data events passing all selections in the mass range used for $H \rightarrow 4\ell$ measurements ($m_{4\ell} > 70$ GeV) are listed in Table ?? for full Run 2 datasets.

Figure ?? shows the reconstructed four-leptons invariant mass distribution for whole Run 2 datasets and MC samples in full mass range and high mass range.

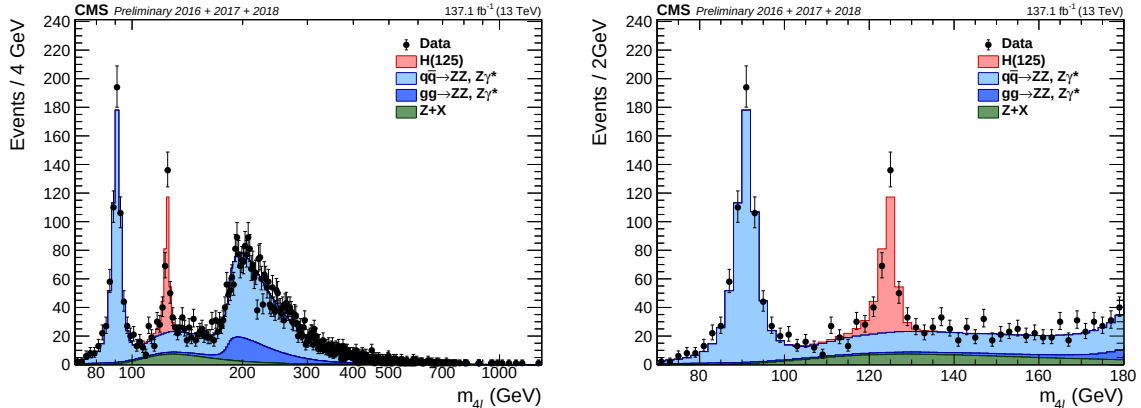


FIGURE 6: Distributions of invariant mass of four-leptons using full Run 2 datasets in the full mass range (left) and the low-mass range (right), along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \rightarrow 4\ell$ resonance.

0.3.4 Definition of the Fiducial Volume

The fraction of all $H \rightarrow 4\ell$ events that get selected following the event selection procedure can change significantly for various Higgs boson production mechanism and distinct exotic models of the Higgs boson. The acceptance difference between SM $H \rightarrow 4\ell$ decays and other exotic models (Higgs-like spin-one models), processes with large hadronic activity (like $t\bar{t}H$ production) and different kinematics processes (like $H \rightarrow Z\gamma^*(\gamma^*\gamma^*) \rightarrow 4\ell$) can vary up to 70%. In order to minimize

the dependence of the measurements on the underlying assumption on the models of the Higgs boson production and properties, the fiducial phase space for the $H \rightarrow 4\ell$ cross section measurements is defined to match as closely as possible the experimental acceptance defined by the reconstruction level selection.

The fiducial phase space demands for at least four leptons having $p_T > 7(5)$ GeV for electrons (muons) with leading and sub-leading leptons having $p_T > 20$ GeV and $p_T > 10$ GeV. The four-momenta of each lepton is calculated at the hard scattering level (PYTHIA status code = 3), before any final state radiation occurs. All electrons (muons) must have pseudorapidity $|\eta| < 2.5(2.4)$. Each lepton isolation is calculated by adding p_T of all stable particles (PYTHIA status code = 1) within a cone of radius $\Delta R < 0.4$ around that lepton. All neutrinos and FSR photons are excluded from above summation. Then the relative isolation of the lepton is defined by the ratio of this sum and p_T of the considered lepton which is required to be less than 0.4, consistent with reconstruction level isolation requirement [Chatrchyan:2013mx]. The requirement of isolation in fiducial phase space definition is crucial to control the difference of signal selection efficiency among different models. The studies based on simulation samples show that signal efficiency can change up to 45% among different models without requiring lepton isolation at fiducial level as shown in Table ??, the maximum difference comes from $t\bar{t}H$ and $gg \rightarrow H$ production modes. This behavior is expected as $t\bar{t}H$ events has more hadronic activity around leptons as compared to $gg \rightarrow H$ events, so the cut at relative isolation kills more events in $t\bar{t}H$ than $gg \rightarrow H$.

Furthermore, fiducial phase space is defined by several other topological and kinematics selections. There must be two SFOS lepton pairs, Z_1 is reconstructed with invariant mass closest to the world average Z boson mass by all possible combination of same flavor and opposite charge (SFOS) pairs of leptons within mass range of $40 \leq m(Z_1) \leq 120$ GeV. Z_2 is reconstructed by remaining SFOS pair of leptons within inv. mass range of $12 \leq m(Z_2) \leq 120$ GeV. If more than one Z_2 passes the criteria, the leptons pair with largest $\Sigma|p_T|$ is selected. To remove the overlapping between leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. To remove contamination from QCD low masses resonance, any opposite-charge pair of leptons is required to pass $m(\ell^+ \ell'^-) \geq 4$ GeV. Finally, We require that all selected leptons must emerge from Higgs boson decay and invariant mass of four-leptons must fulfill $105 \text{ GeV} < m(4\ell) < 140 \text{ GeV}$. The demand for $m(4\ell)$ helps to control the contamination from off-shell gg fusion production which is of considerable size, capable to increase the total cross section by a few percent [Kauer:2013cga, Kauer:2012hd]. The requirements that defines fiducial phase space are also gathered in Table 1.

It has been established based on simulation samples that with including isolation for the events passing fiducial phase space definition reconstruction efficiencies faintly depends on Higgs boson properties and production modes. The remaining model dependence will be quoted as a separate uncertainty as explained in Section ?. Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and reconstruction efficiencies (ϵ) for SM and exotic production modes are registered in Tables 2 and 3.

TABLE 2: Overview of all requirements utilized to define the fiducial phase space for the $H \rightarrow 4\ell$ cross section measurements.

Requirements for the $H \rightarrow 4\ell$ fiducial phase space	
Lepton kinematics and isolation	
Leading lepton p_T	$p_T > 20$ GeV
Sub-leading lepton p_T	$p_T > 10$ GeV
Additional electrons (muons) p_T	$p_T > 7$ (5) GeV
Pseudorapidity of electrons (muons)	$ \eta < 2.5$ (2.4)
Sum of scalar p_T of all stable particles within $\Delta R < 0.4$ from lepton	$< 0.4 p_T$
Event topology	
Existence of at least two SFOS lepton pairs, where leptons satisfy criteria above	
Inv. mass of the Z_1 candidate	$40 < m(Z_1) < 120$ GeV
Inv. mass of the Z_2 candidate	$12 < m(Z_2) < 120$ GeV
Distance between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$
Inv. mass of any opposite-sign lepton pair	$m(\ell_i^+ \ell_j^-) > 4$ GeV
Inv. mass of the selected four leptons	$105 < m_{4\ell} < 140$ GeV

TABLE 3: The fiducial acceptance ($\mathcal{A}_{\text{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 8 TeV signal models assuming $m_H = 125.0$ GeV, the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
$gg \rightarrow H$ (POWHEG+JHUGEN)	0.422 ± 0.001	0.647 ± 0.002	0.053 ± 0.001	0.681 ± 0.002
$gg \rightarrow H$ (MINLOHJJ)	0.415 ± 0.001	0.636 ± 0.002	0.051 ± 0.001	0.668 ± 0.002
VBF (POWHEG)	0.476 ± 0.003	0.652 ± 0.005	0.040 ± 0.002	0.678 ± 0.005
WH (PYTHIA)	0.342 ± 0.002	0.627 ± 0.003	0.072 ± 0.002	0.672 ± 0.003
ZH (PYTHIA)	0.348 ± 0.003	0.634 ± 0.004	0.072 ± 0.003	0.679 ± 0.005
$t\bar{t}H$ (PYTHIA)	0.250 ± 0.003	0.601 ± 0.008	0.139 ± 0.008	0.685 ± 0.010
Some characteristic models of Higgs-like boson with exotic decays and properties				
$q\bar{q} \rightarrow H(J^{\text{CP}}=1^-)$ (JHUGEN)	0.238 ± 0.001	0.609 ± 0.002	0.054 ± 0.001	0.642 ± 0.002
$q\bar{q} \rightarrow H(J^{\text{CP}}=1^+)$ (JHUGEN)	0.283 ± 0.001	0.619 ± 0.002	0.051 ± 0.001	0.651 ± 0.002
$gg \rightarrow H \rightarrow Z\gamma^*$ (JHUGEN)	0.156 ± 0.001	0.622 ± 0.002	0.073 ± 0.001	0.667 ± 0.002
$gg \rightarrow H \rightarrow \gamma^*\gamma^*$ (JHUGEN)	0.188 ± 0.001	0.629 ± 0.002	0.066 ± 0.001	0.671 ± 0.002

TABLE 4: The fiducial acceptance ($\mathcal{A}_{\text{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Valued are obtained using 7 TeV signal models assuming $m_H = 125.0$ GeV, the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
$gg \rightarrow H$ (POWHEG+JHUGEN)	0.438 ± 0.001	0.676 ± 0.002	0.049 ± 0.001	0.709 ± 0.002
$gg \rightarrow H$ (MINLOHJJ)	0.427 ± 0.001	0.675 ± 0.002	0.047 ± 0.001	0.707 ± 0.002
VBF(POWHEG)	0.483 ± 0.003	0.674 ± 0.004	0.035 ± 0.002	0.697 ± 0.005
WH(PYTHIA)	0.357 ± 0.002	0.660 ± 0.004	0.061 ± 0.002	0.700 ± 0.050
ZH(PYTHIA)	0.372 ± 0.004	0.662 ± 0.006	0.059 ± 0.004	0.701 ± 0.007
$t\bar{t}H$ (PYTHIA)	0.248 ± 0.004	0.636 ± 0.010	0.158 ± 0.011	0.736 ± 0.013
Some characteristic models of Higgs-like boson with exotic decays and properties				
$q\bar{q} \rightarrow H(J^{CP} = 1^-)$ (JHUGEN)	0.254 ± 0.001	0.645 ± 0.002	0.049 ± 0.001	0.677 ± 0.002
$q\bar{q} \rightarrow H(J^{CP} = 1^+)$ (JHUGEN)	0.301 ± 0.001	0.657 ± 0.002	0.044 ± 0.001	0.686 ± 0.002
$gg \rightarrow H \rightarrow Z\gamma^*$ (JHUGEN)	0.160 ± 0.001	0.654 ± 0.002	0.072 ± 0.001	0.701 ± 0.002
$gg \rightarrow H \rightarrow \gamma^*\gamma^*$ (JHUGEN)	0.185 ± 0.001	0.659 ± 0.002	0.064 ± 0.001	0.701 ± 0.002

TABLE 5: The fiducial acceptance ($\mathcal{A}_{\text{fid}}$) per final state using 8 TeV signal samples.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
$gg \rightarrow H$ (POWHEG+JHUGEN) 125 GeV	0.409 ± 0.003	0.453 ± 0.003	0.413 ± 0.002	0.422 ± 0.001
$gg \rightarrow H$ (POWHEG+JHUGEN) 126 GeV	0.410 ± 0.003	0.460 ± 0.003	0.422 ± 0.002	0.429 ± 0.001
$gg \rightarrow H$ (POWHEG) 125 GeV	0.396 ± 0.003	0.437 ± 0.003	0.411 ± 0.002	0.413 ± 0.001
$gg \rightarrow H$ (POWHEG) 126 GeV	0.402 ± 0.003	0.447 ± 0.003	0.418 ± 0.002	0.421 ± 0.001
$gg \rightarrow H$ (POWHEG1.0) 125 GeV	0.395 ± 0.001	0.443 ± 0.001	0.411 ± 0.001	0.415 ± 0.001
$gg \rightarrow H$ (POWHEG1.0) 126 GeV	0.408 ± 0.003	0.446 ± 0.003	0.421 ± 0.002	0.424 ± 0.001
$gg \rightarrow H$ (MINLOHJJ) 125 GeV	0.394 ± 0.002	0.439 ± 0.002	0.413 ± 0.002	0.415 ± 0.001
$gg \rightarrow H$ (MINLOHJJ) 126 GeV	0.407 ± 0.002	0.449 ± 0.002	0.418 ± 0.002	0.423 ± 0.001
VBF (POWHEG) 125 GeV	0.456 ± 0.007	0.503 ± 0.007	0.473 ± 0.005	0.476 ± 0.003
VBF (POWHEG) 126 GeV	0.450 ± 0.007	0.513 ± 0.007	0.471 ± 0.005	0.476 ± 0.003
WH (PYTHIA) 125 GeV	0.332 ± 0.003	0.365 ± 0.003	0.336 ± 0.002	0.342 ± 0.002
WH (PYTHIA) 126 GeV	0.331 ± 0.003	0.365 ± 0.003	0.340 ± 0.002	0.344 ± 0.002
ZH (PYTHIA) 125 GeV	0.343 ± 0.005	0.367 ± 0.005	0.341 ± 0.004	0.348 ± 0.003
ZH (PYTHIA) 126 GeV	0.339 ± 0.005	0.384 ± 0.005	0.362 ± 0.004	0.362 ± 0.003
$t\bar{t}H$ (PYTHIA) 125 GeV	0.239 ± 0.007	0.274 ± 0.007	0.244 ± 0.005	0.250 ± 0.003
$t\bar{t}H$ (PYTHIA) 126 GeV	0.261 ± 0.007	0.271 ± 0.007	0.255 ± 0.005	0.260 ± 0.004